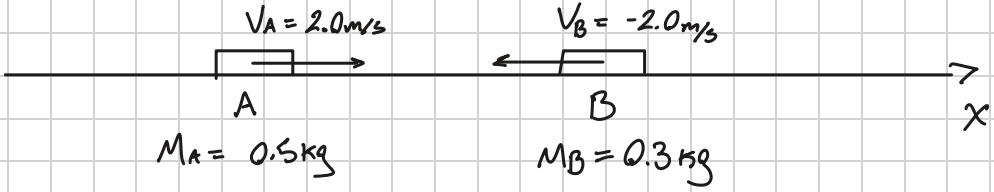
week 9

6.4 elastic collision

Example



Assuming a perfectly elastic collision what are the velocities after?

$$M_A V_{A_1} + M_B V_{B_1} = M_A V_{A_2} + M_B V_{B_2} \quad (1)$$

$$V_{B_2} - V_{A_2} = -(V_{B_1} - V_{A_1}) \quad (2)$$

plug in numbers in (2) ⟶

$$V_{B_2} - V_{A_2} = -(-2.0 - 2.0)$$

$$\Rightarrow V_{B_2} = V_{A_2} + 4.0 m/s$$

insert into equation (1)

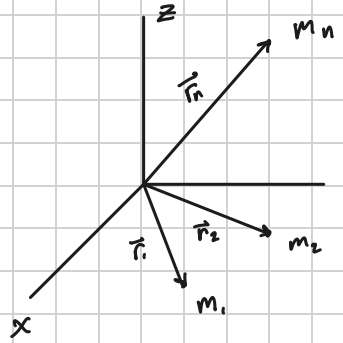
8.5 the center of mass

System of n point particles of mass $m_1, m_2, m_3, m \ldots m_n$

each has position $(x_1, y_1, z_1), (x_2, y_2, z_2) \ldots (x_n, y_n, z_n)$

Def center of mass of the system is defined by:

$$X_{cm} = \frac{m_1x_1 + m_2x_2 + \cdots + m_nx_n}{m_1 + m_2 + \ldots + m_n} \rightarrow \text{same for } y \text{ and } z$$



$$\vec{r}_{cm} = \frac{m_1r_1 + m_2r_2 + \cdots m_nr_n}{m_1 + m_2 + \cdots + m_n}$$

$M = m_1$

Whoops got distracted

take $\frac{d}{dt}$ on both sides

$$\frac{d}{dt}\vec{r}_{cm} = \frac{d}{dt}\frac{\sum m_i \vec{r}_i}{M}$$

$$\Rightarrow \vec{V}_{cm} = \frac{m_1 v_1 + m_2 v_2 + \cdots + m_n v_n}{M}$$

$$\Rightarrow M\vec{v}_{cm} = m_1 v_1 + m_2 v_2 + \cdots + m_n v_n$$

$$= p_1 + p_2 + \cdots + p_n$$

$$= \vec{p}_{tot}$$

Looks as if the entire mass of the system is localized at r_{cm} $\vec{p}_{tot} = M \cdot \vec{V}_{cm} \Rightarrow \vec{V}_{cm} = \frac{\vec{p}_{tot}}{M}$

Apply $\frac{d}{dt}$ again: ↓

$$\frac{d}{dt}\left(M\vec{V}_{cm} = m_1\vec{V}_1 + m_2\vec{V}_2 + \cdots + m_n\vec{V}_n\right)$$

$$M\vec{a}_{cm} = m_1\vec{a}_1 + \cdots + m_n\vec{a}_n$$

for each particle we wright newtons seeard law

$$m_1\vec{a}_1 = \vec{F}_{ext} + \vec{F}_{int}$$

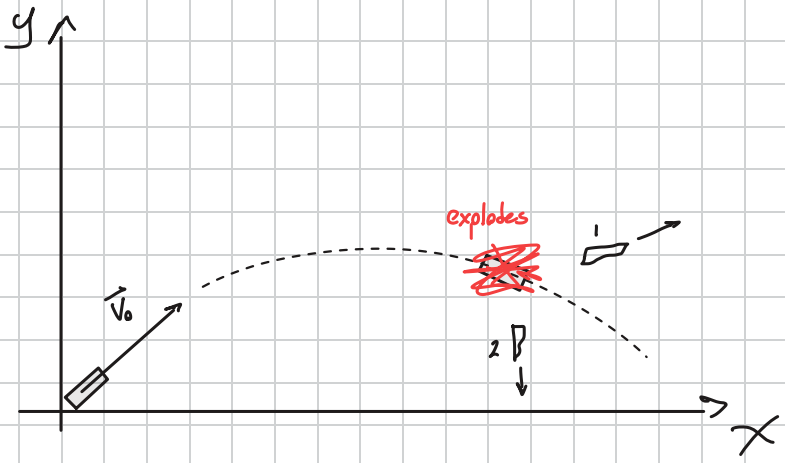
$$= \sum \vec{F}_{ext} + \cancel{\sum \vec{F}_{int}}$$

newtons third law

$$\sum \vec{F}_{ext} = \vec{F}_{ext}$$

$$\Rightarrow \vec{F}_{ext} = M\vec{a}_{cm}$$

Example



trajectory of center of mass is undisturbed

External forces : gravity only

internal forces : Cause breaks up of the projectile

Always $\sum \vec{F}_{int} = 0 \Rightarrow M\vec{a}_{cm} = \vec{F}_{ext}$